

Investigating strategies for next-generation flavivirus vaccine development

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ABSTRACT

Dengue virus (DV) is the most common mosquito-borne virus in the world, with 4 billion people at risk for infection by any of its four serotypes (DV1-4). Severe cases of DV and related flavivirus Zika virus (ZV) are associated with antibody-dependent enhancement (ADE) of infections. Currently, there is a need for vaccines that do not carry the risk of inducing ADE. Neutralizing antibodies against DV and ZV have shown to target domain III (DIII) of E glycoprotein, which is critical for viral entry and cell attachment, with broadly protective antibodies targeting the DIII lateral ridge (LR). We previously demonstrated strong immunoprotecting potential of a homotypic ZV immunization strategy, which involved immunization with nanoparticles decorated with LR-focused ZVDIII mutant. We are studying the effectiveness of heterotypic immunization strategies against ZV and DV and characterizing humoral responses to these immunizations in mice. These studies will further our understanding of the adaptive immune response to flavivirus infections and may provide a framework for developing novel vaccines and antibody-based therapeutics.

BACKGROUND

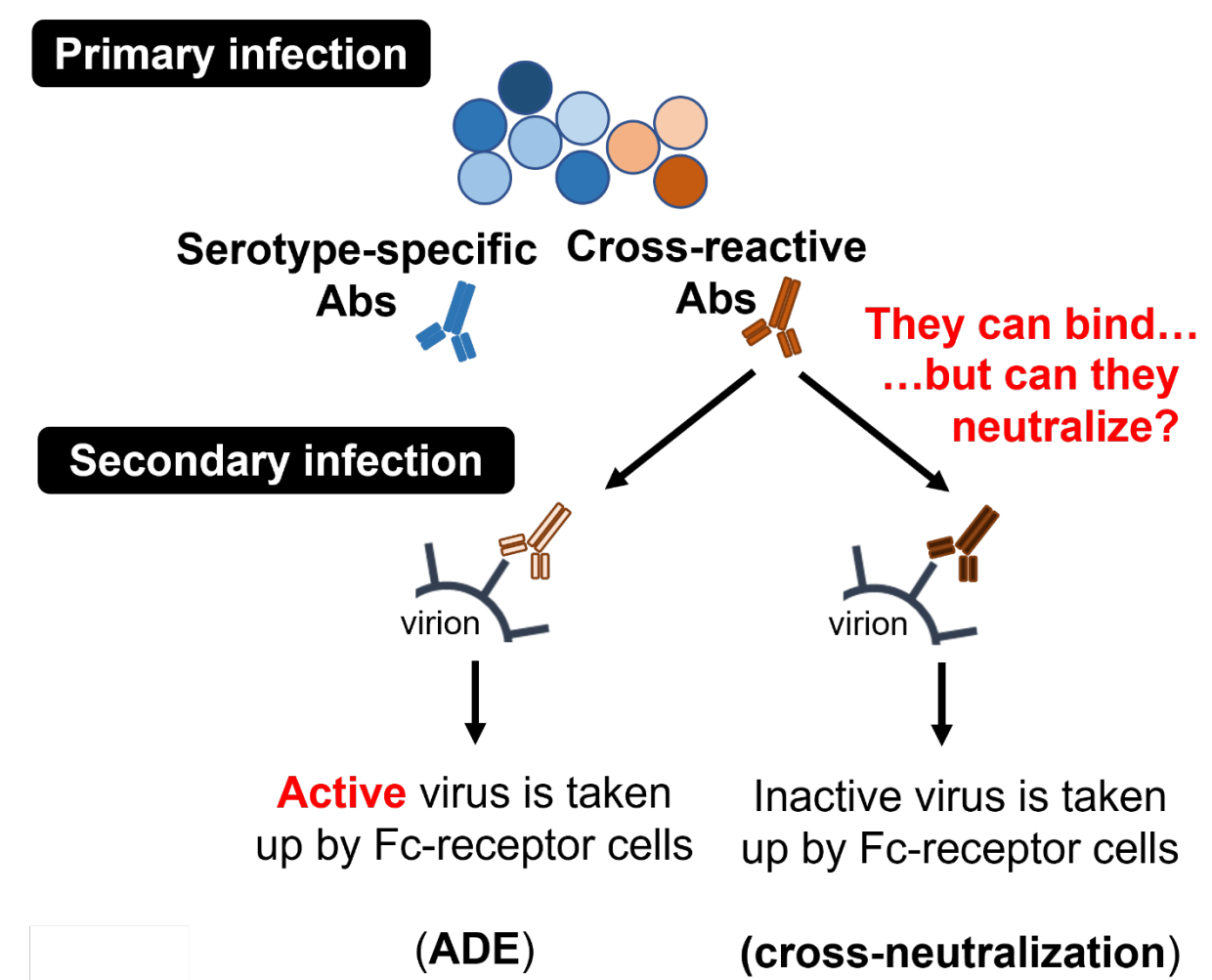


Figure 1. Proposed mechanism of antibody-dependent enhancement (ADE). Phenomenon occurs due to interactions between flavivirus infections.

- ADE interactions of DV 1-4 and ZV
 - Protective antibodies from primary infection
 - During secondary infection, they may be non- or weakly-neutralizing, instead increasing viremia and levels of pro-inflammatory cytokines

- E glycoprotein mediates cell attachment and viral entry
- LR region of DIII: A-strand, BC-loop, DE-loop, FG-loop
- LR targeted by broadly neutralizing antibodies (bNAbs) across DV and ZV

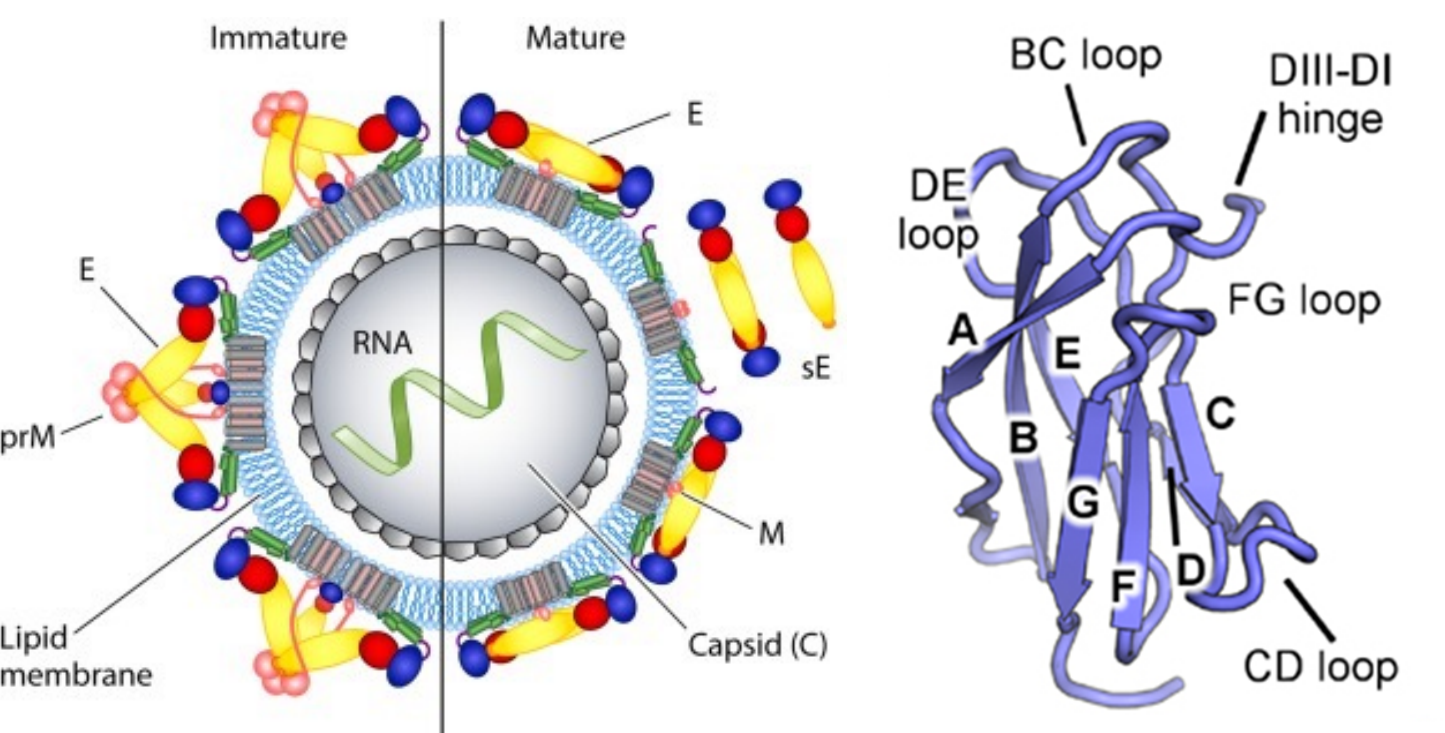


Figure 2. Flavivirus structural proteins. A) Proteins on the surface of immature and mature virions.¹ B) Domain III (DIII) of E glycoprotein and its motifs.²

- Immunization
 - Prime the body to circulate pre-infection bNAbs
 - To mitigate severity and protect against infection without the risk of ADE

BACKGROUND (cont'd)

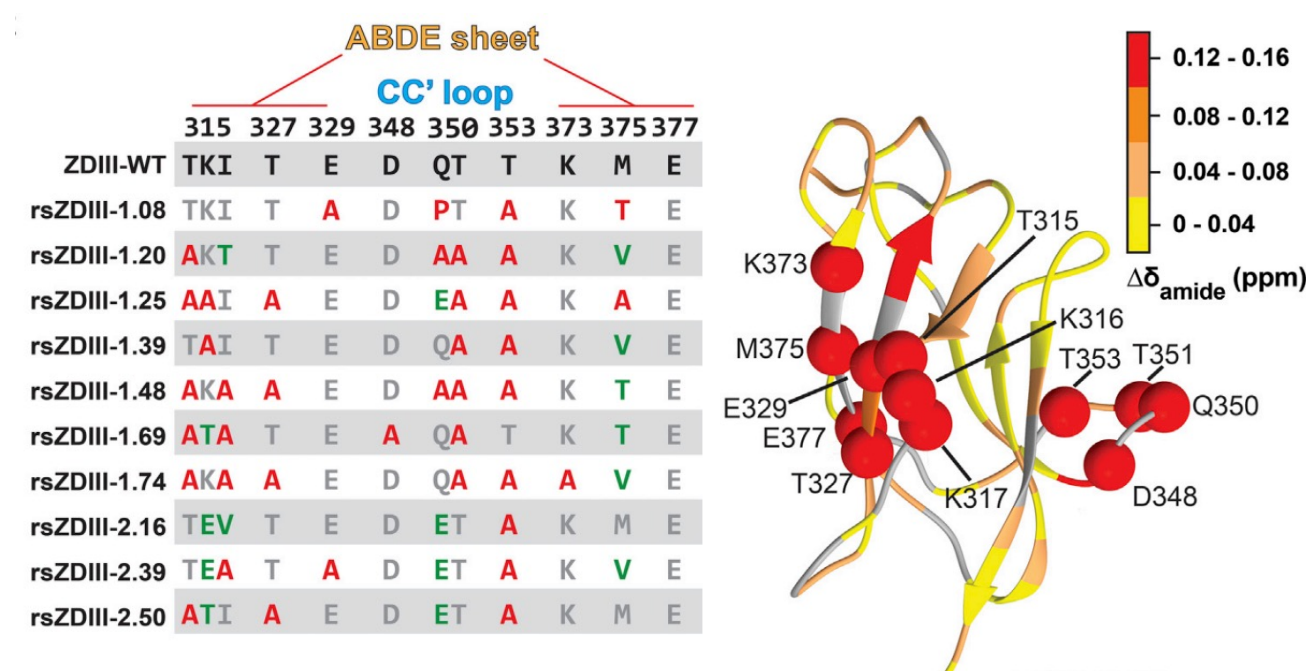


Figure 3. Characterization of rsZVDIII.3. (L) Shortlist of candidates. (R) Chemical shift differences of rsZVDIII-2.39 presented on structural model of wtZVDIII (PDB: 5KVG).

- Resurfaced ZV DIII candidates (rsZVDIII) conjugated onto Lumazine Synthase nanoparticles (LS; NPs) for multivalent immunogen display

- Immunized mice with various homotypic rsZVDIII NPs formats

- Sera of rsZVDIII-2.39 NP demonstrated protective antibody titers against ZV

- Sera exhibited cross-reactivity across DV 1, 3, and 4 but lacked cross-neutralization activity

- Focus B cell response against LR by dampening response against other DIII epitopes

- Masked epitopes with mutations via phage display, generating *resurfaced DIIIs* (rsDIIIs)

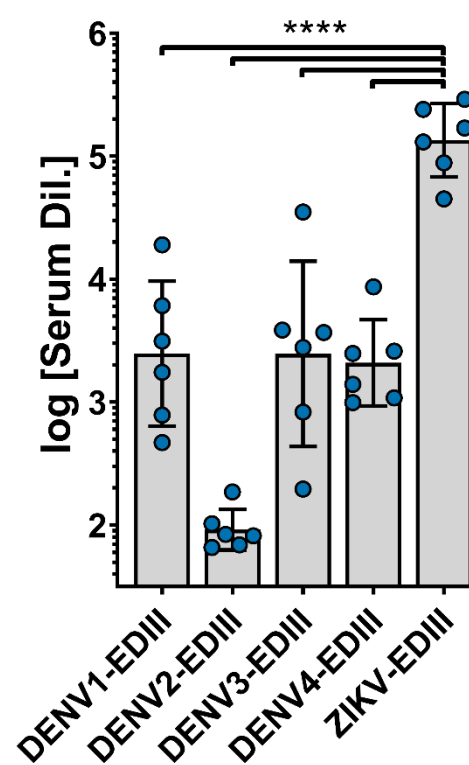


Figure 4. ELISA reactivities of rsZVDIII-2.39 NP-immunized sera. Cross-reactivity across DV DIIIs is reserved.

RESULTS

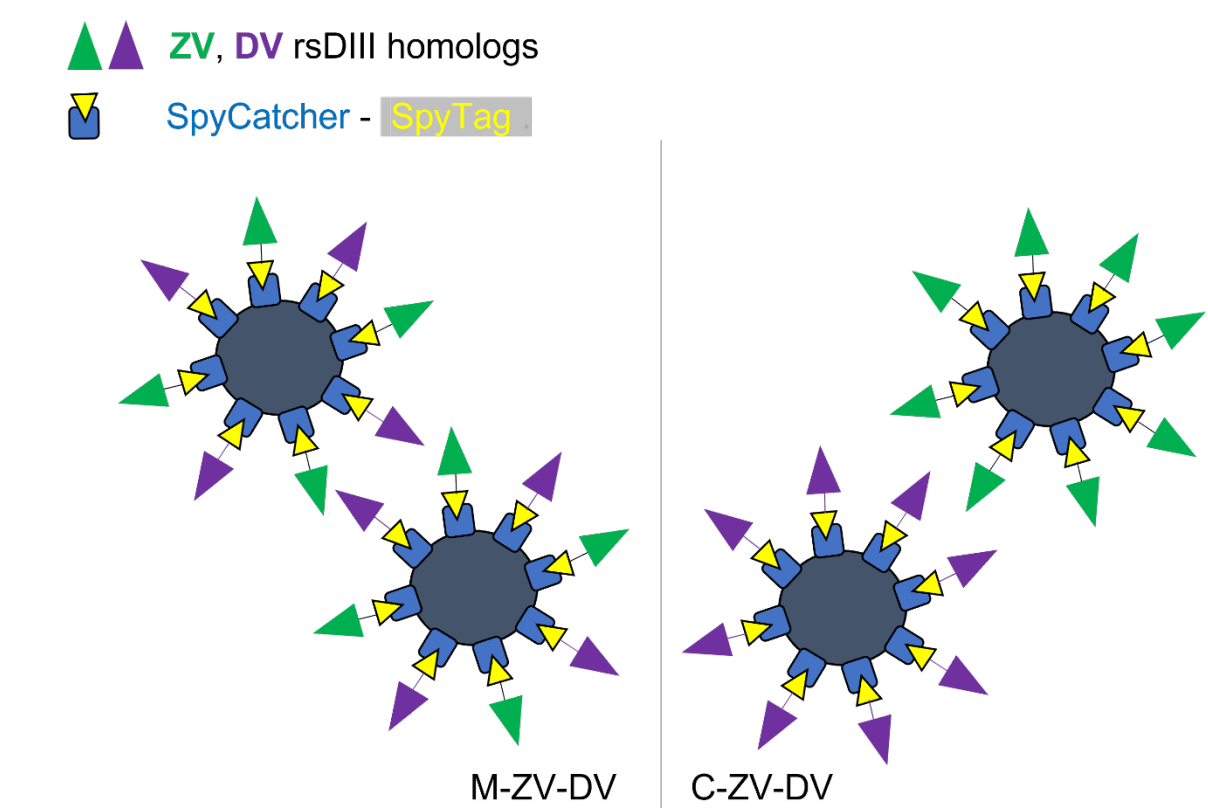


Figure 5. Designs of hetero-NP DIII delivery. Mosaic design (M-ZV-DV) involves conjugating both rsDIIIs onto each NP. Cocktail design (C-ZV-DV) involves a mixture of NPs, each conjugated to one rsDIII homolog.

- Conjugation using SpyT/SpyCatcher, a spontaneous, irreversible reaction

- LS derived from *Aquifex aeolius*
 - His-tagged, purified with Ni-NTA chromatography

- DIII immunogens
 - His-tagged and fused to maltose binding protein
 - Separated MBP from DIIIs with TEV protease, purified with anion-exchange chromatography

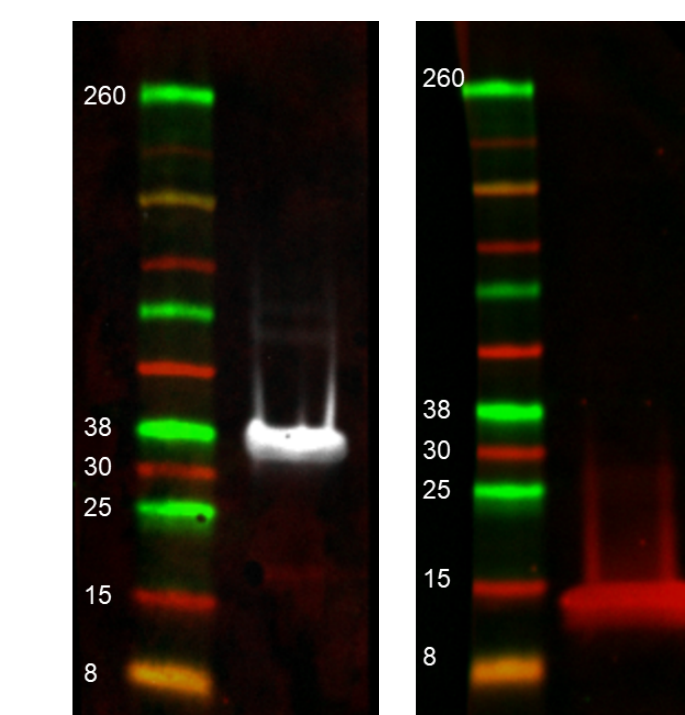


Figure 6. Purified NP components. Western blots of (L) LS stained with anti-His-HRP and (R) wtDV1DIII-SpyT with Alexa Fluor 680.

RESULTS (cont'd)

- Introduced LS to rsZVDIII and wtDV1DIII in 1.5 molar excess

- Allowed SpyT/SpyC conjugation reaction for 24 hours

- Purified with size exclusion chromatography

- Preparing M-rsZV-wtDV1, rsZV, and wtDV1 NP formats

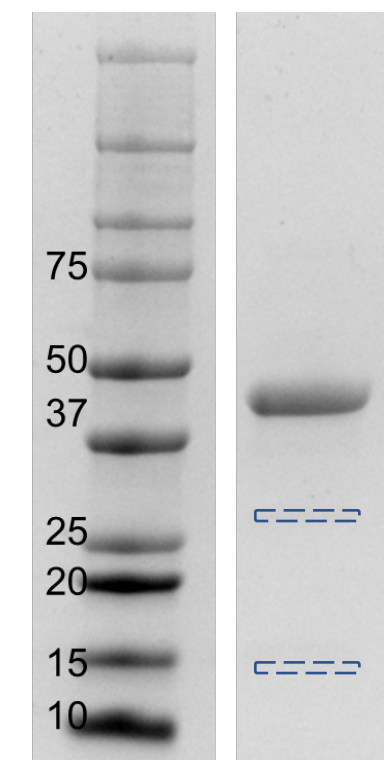


Figure 7. Final product after NP assembly reactions. Image of purified protein run on SDS-PAGE gel stained with Coomassie blue. Free proteins would run at the following dashed bands.

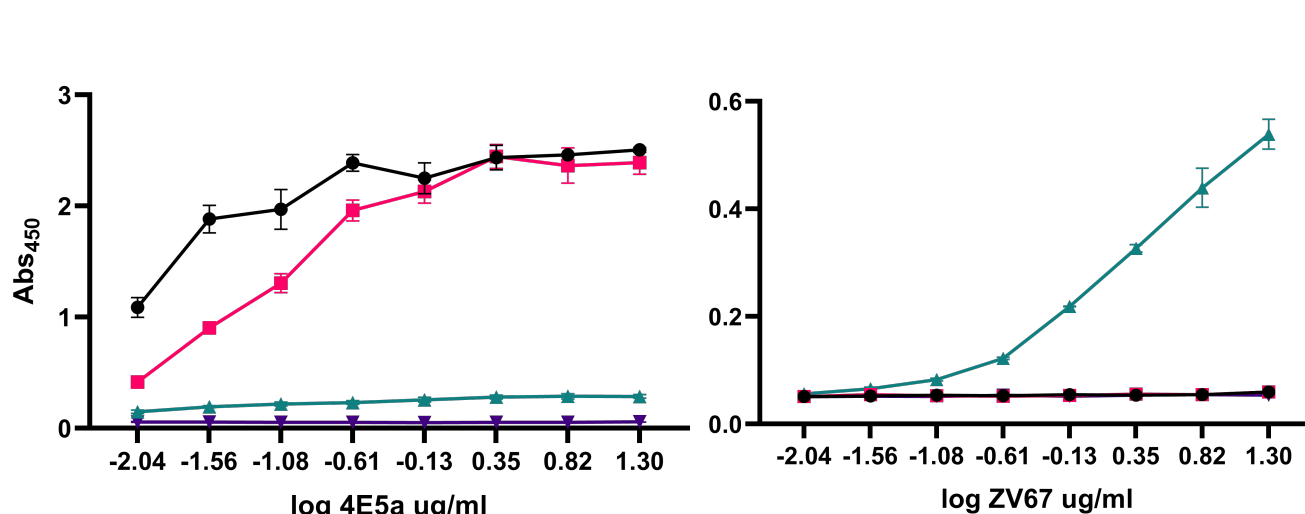
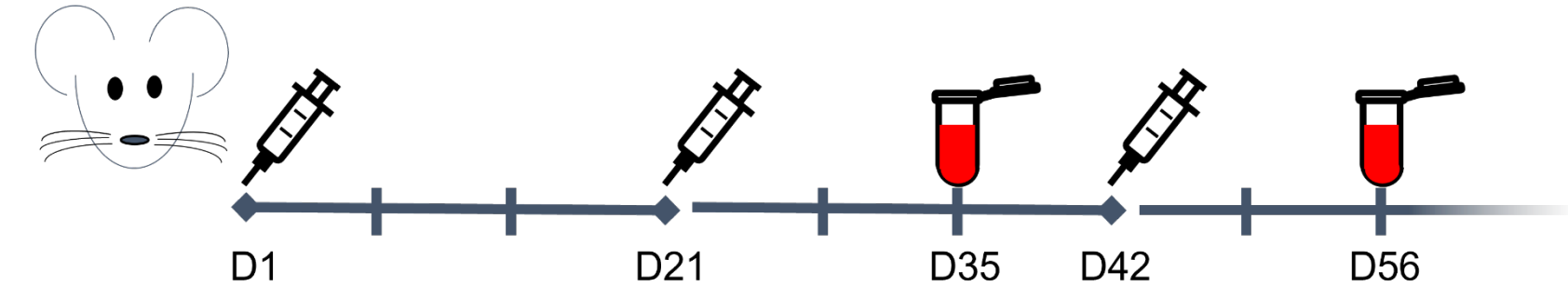


Figure 8. ELISA reactivities against LR-binding antibodies. 4E5a binds to DV DIII and ZV67 to ZVDIII. Black, wtDV1DIII NPs; blue, rsZVDIII 2.39 NPs; magenta, wtDV1DIII-SpyT only; purple, BSA only.

- Characterized homotypic NP formats via ELISA reactivity profiles against LR-binding antibodies

- Administer NP formats, with adjuvant Addavax, to 6-week-old BALB/c mice
 - Intraperitoneal injections

Figure 9. Three-part immunization procedure. Injections are three weeks apart. Submandibular bleeds are two weeks after each booster.



FUTURE DIRECTIONS

- Determine molar relationship between immunogens in M-rsZV-wtDV1
- Study the neutralization activity of sera antibodies
- Employ viral challenge studies in mice and other studies evaluating ADE activity following mosaic NP immunization
- Investigate the immunoprotecting potential of other heterotypic NP formats, such as cocktail NPs
- Characterize differences in elicited antibody response between mosaic and cocktail NP formats, using methods such as RNA-sequencing of resulting B cell receptors

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